MAPLE: Manifold-Activated Pruning of Latent Experts for Sub-2-BPW Mixture-of-Experts Language Models

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***Abstract*—We present MAPLE (Manifold-Activated Pruning of Latent Experts), a post-training structural compression methodology for fine-grained Mixture-of-Experts (MoE) large language models that achieves sub-2-bit-per-weight effective parameter density while retaining high task-level semantic fidelity. While global multilingual pre-training necessitates massive expert populations (e.g., 128–256 experts), typical edge deployment domains draw upon a surprisingly compact manifold of routed expert capacity. We formalize this sparsity by profiling empirical forward-pass expert activations over a representative domain corpus. Each expert is evaluated via a composite metric combining activation frequency and input feature energy: $s_{e,l} = C_{e,l} \cdot \sqrt{\bar{\sigma}_{e,l} + \epsilon}$. Applied to Gemma 4 26B-A4B (128 routed experts, 30 layers, 9.92 GB in IQ2_XXS quantization), MAPLE excises 80 dormant experts per layer to assemble an optimized 48-expert container of 4.75 GB (effective 1.48 bits per parameter, 2.09× compression). The compressed model achieves 100% GPU VRAM residency on a consumer 6 GB laptop GPU (RTX 4050), accelerating generation by 6.57× from 6.9 to 45.4 tokens/second by eliminating PCIe memory-paging bottlenecks. Standardized task evaluation across code synthesis, data structure implementation, scientific deduction, and formal logic demonstrates a 93.8% mean keyword fidelity retention. We document three subtle failure modes overcome during design: proxy-embedding routing estimation, cascaded normalization suppression, and DRY sampling interference.**

***Index Terms*—Mixture of Experts, Model Compression, Edge Inference, Structured Pruning, GGUF, Quantization, High-Throughput Systems.**

I. Introduction

Mixture-of-Experts (MoE) architectures [1]–[3] have become the foundational paradigm for scaling modern large language models, decoupling total parameter capacity from per-token compute by routing inputs through a sparse subset of k active experts out of E total candidates ($k \ll E$). While this design maintains sub-linear FLOP counts during generation, running frontier MoE models on consumer edge hardware remains fundamentally bottlenecked by GPU Video RAM (VRAM) capacity.

Even when quantized to extreme low-bit representations such as 2-bit quantization (e.g., Gemma 4 26B-A4B at IQ2_XXS), the resulting model footprint exceeds 9.92 GB. On consumer laptops and workstations equipped with common 6 GB or 8 GB GPUs (such as the NVIDIA RTX 4050 or RTX 3060), the model cannot fit into high-bandwidth VRAM. Consequently, inference runtimes must offload layers to system RAM over the PCIe bus. This offloading creates severe memory-bandwidth stalls, causing throughput to collapse from interactive conversational speeds to single digits (6.9 tokens/second on an RTX 4050 Laptop GPU). In edge computing, achieving 100% in-VRAM execution represents a discrete operational cliff: crossing below the physical VRAM threshold transforms a model from an unusable system into a real-time local copilot.

Existing compression literature is largely bifurcated into:

1) *Numerical Quantization:* Techniques such as GPTQ [4], AWQ [5], and SpQR [6] compress weights from 16-bit floats to 4-bit, 3-bit, or 2-bit integers. However, pushing numerical precision below 2 bits per weight yields steep quantization noise, gradient instability, and catastrophic perplexity explosion.
2) *Structural Pruning:* Unstructured and semi-structured weight pruning methods (e.g., SparseGPT [7]) typically require specialized sparse matrix multiplication kernels that yield minimal hardware acceleration on consumer GPUs without custom tensor cores.

Fine-grained MoE models provide an orthogonal, unexplored dimension for compression: *expert capacity allocation*. During pre-training, an expansive expert pool ($E = 64$–128) is required to capture hundreds of human languages, diverse programming dialects, and encyclopedic facts. However, edge deployments are rarely omni-domain. In practice, a local deployment is dedicated to a specific operational subdomain, such as English technical reasoning, algorithmic development, and conversational assistance. Within this domain, a substantial fraction of experts remain completely unselected.

In this paper, we introduce **MAPLE** (**M**anifold-**A**ctivated **P**runing of **L**atent **E**xperts), an empirical forward-pass profiling and structural pruning framework designed specifically for fine-grained MoE architectures in the GGUF ecosystem [11]. MAPLE captures the empirical activation manifold of the target domain, computes a joint frequency-energy saliency score for every expert across all layers, and directly excises dormant experts from quantized GGUF binaries without requiring retraining, gradient access, or dequantization.

Applied to Gemma 4 26B-A4B, MAPLE compresses the model from 9.92 GB to 4.75 GB—achieving an effective density of **1.48 bits per parameter** (a 2.09× reduction over an already 2-bit quantized base). Crucially, the 4.75 GB model fits

entirely within the 6141 MiB VRAM budget of an RTX 4050 Laptop GPU, accelerating generation by **6.57× (from 6.9 to 45.4 TPS)** while retaining **93.8% semantic task fidelity**.

II. The MAPLE Framework

A. MoE Architectural Mechanics

In a standard MoE transformer block, the conventional feed-forward network (FFN) is replaced with a routing gate and E parallel expert networks. Given an incoming hidden state $\mathbf{h} \in \mathbb{R}^d$, the router computes unnormalized logits $\mathbf{g} = \mathbf{h}W_{\text{gate}}$, where $W_{\text{gate}} \in \mathbb{R}^{d \times E}$. The top-$k$ experts with the highest routing scores are selected:

$$\mathcal{T}_k(\mathbf{h}) = \text{TopK}(\mathbf{g}, k) \tag{1}$$

The gate probabilities are computed via a softmax over the selected top-k logits:

$$p_e(\mathbf{h}) = \frac{\exp(g_e)}{\sum_{j \in \mathcal{T}_k(\mathbf{h})} \exp(g_j)}, \quad \forall e \in \mathcal{T}_k(\mathbf{h}) \tag{2}$$

The block output is the weighted sum of the active experts' outputs plus any shared dense projection:

$$\mathbf{y} = \text{FFN}_{\text{dense}}(\mathbf{h}) + \sum_{e \in \mathcal{T}_k(\mathbf{h})} p_e(\mathbf{h}) \cdot \text{FFN}_e(\mathbf{h}) \tag{3}$$

In Gemma 4 26B-A4B, $E = 128$ routed experts exist per block across $L = 30$ layers, with $k = 8$ experts active per token. The routed expert weights (`ffn_down_exps` and `ffn_gate_up_exps`) constitute 8.24 GB of the total 9.92 GB container size.

B. Domain Calibration Corpus

To discover which experts form the operative manifold, we construct a calibration dataset $\mathcal{D}_{\text{calib}}$ covering the target deployment distribution. For our implementation, we assembled a 1,979-token multi-domain corpus structured under the model's native conversational template (`<|turn>user ... <turn|> <|turn>model`):

- **Algorithmic Code Synthesis (30%):** Recursive functions, dynamic programming, sorting routines, binary search trees, and Python decorators.
- **Mathematical Deduction (20%):** Matrix algebra, combinatorial probability, and continuous optimization.
- **Scientific Reasoning (20%):** Thermodynamics, Rayleigh scattering, wave mechanics, and chemical kinetics.
- **Formal Deductive Logic (15%):** Syllogisms, transitive inference chains, and set inclusion.
- **Structured Discourse (15%):** Technical summarization and precise instruction adherence.

C. Empirical Forward-Pass Profiling

Rather than approximating expert importance using static weight statistics, MAPLE captures the live forward-pass activation statistics using `llama-imatrix`. As tokens traverse the network, the profiler tracks two empirical tensors for each layer $l \in [0, L-1]$:

1) **Activation Frequency Vector ($C_{e,l} \in \mathbb{N}^E$):** The exact count of tokens routed to expert e:

$$C_{e,l} = \sum_{\tau \in \mathcal{D}_{\text{calib}}} \mathbb{I}(e \in \mathcal{T}_k(\mathbf{h}_{\tau,l})) \tag{4}$$

2) **Input Energy Sum Matrix ($\Sigma_{e,l} \in \mathbb{R}^{E \times d_{\text{ff}}}$):** The accumulated squared magnitude of the input features across all $d_{\text{ff}} = 704$ intermediate projection channels:

$$\Sigma_{e,l,j} = \sum_{\tau \in \mathcal{D}_{\text{calib}}: e \in \mathcal{T}_k(\mathbf{h}_{\tau,l})} (h_{\tau,l,j})^2 \tag{5}$$

From the energy sum, we compute the mean channel energy:

$$\bar{\sigma}_{e,l} = \frac{1}{d_{\text{ff}}} \sum_{j=1}^{d_{\text{ff}}} \Sigma_{e,l,j} \tag{6}$$

D. The MAPLE Composite Saliency Score

A naive frequency count $C_{e,l}$ ignores the magnitude of the signal processed by the expert: an expert called frequently on low-norm padding tokens may contribute minimally to the residual stream. Conversely, sorting purely by raw energy $\bar{\sigma}_{e,l}$ is fragile against stochastic outliers.

MAPLE defines a balanced composite scoring function:

$$s_{e,l} = C_{e,l} \cdot \sqrt{\bar{\sigma}_{e,l} + \epsilon} \tag{7}$$

where $\epsilon = 10^{-8}$ prevents numerical instability.

For each layer l, the $E = 128$ experts are sorted by $s_{e,l}$ descending, and the top $k' = 48$ expert indices are extracted into a layer selection map $\mathcal{S}_l \subset \{0, \ldots, 127\}$ with $|\mathcal{S}_l| = 48$.

E. Empirical Activation Mass Coverage

We define the layer-wise empirical activation coverage as:

$$\text{Coverage}(\mathcal{S}_l) = \frac{\sum_{e \in \mathcal{S}_l} C_{e,l}}{\sum_{e=0}^{E-1} C_{e,l}} \times 100\% \tag{8}$$

Empirically, selecting $k' = 48$ experts captures an average of **79.88%** of the total forward-pass activation mass across all 30 layers (ranging from 71.2% to 92.4% per layer). The discarded 80 experts account for 62.5% of the expert parameter volume but contribute only 20.12% of forward activations on the deployment distribution.

F. Zero-Dequantization Binary Slicing

To assemble the target model without introducing quantization noise or requiring retraining, MAPLE streams the source GGUF binary and performs direct axis-0 index slicing on the quantized tensors:

1) `blk.{l}.ffn_gate_inp.weight`: The router gate matrix $[E, d]$ is sliced along axis 0 to $[\mathcal{S}_l, :]$, yielding shape $[48, d]$.
2) `blk.{l}.ffn_down_exps.weight`: Sliced along expert axis 0 to retain $[\mathcal{S}_l, \ldots]$ in native IQ2_XXS format.
3) `blk.{l}.ffn_gate_up_exps.weight`: Sliced along expert axis 0 to retain $[\mathcal{S}_l, \ldots]$.

4) `blk.{l}.ffn_down_exps.scale`: Quantization scales are sliced to $[\mathcal{S}_l, \ldots]$ with scale multipliers locked at $1.0\times$.
5) **Invariant Tensors:** Attention query, key, value, and output matrices, input/output embeddings, normalization layers, and shared dense FFN blocks are copied directly without byte modification.

III. Analysis of Failure Modes

Achieving high semantic fidelity at 1.48 BPW required resolving three distinct failure modes encountered during development. We document these mechanisms to prevent recurrent pitfalls in MoE pruning.

A. Failure Mode 1: The Ghost Router (Proxy-Embedding)

Symptom: Repetitive token generation loops (e.g., infinite sequences of `nums_nums_nums...`) and total inability to complete logical syllogisms.

Root Cause: An early attempt bypassed forward passes by estimating routing logits from the static token embedding matrix W_{emb}. The mean calibration embedding $\bar{\mathbf{e}}_{\text{calib}}$ was projected through each layer's gate: $\hat{\mathbf{g}}^{(l)} = \bar{\mathbf{e}}_{\text{calib}} W_{\text{gate}}^{(l)}$.

While $\hat{\mathbf{g}}^{(l)}$ provided reasonable alignment in layers 0–2, it diverged completely in deeper layers ($l \geq 10$). In a 30-layer transformer, hidden representations undergo non-linear rotations and multi-head attention projections that shift their geometric trajectory into subspaces orthogonal to W_{emb}. Static embedding projections misidentified 75% of active experts in layers 15–29. Pruning based on this proxy removed active deep experts and preserved dormant ones.

Remedy: Mandatory forward-pass profiling via `llama-imatrix`, ensuring expert selection reflects true hidden-state routing.

B. Failure Mode 2: Cascaded Normalization Suppression

Symptom: Syntactically formed prose with complete factual emptiness and catastrophic loss of reasoning ability.

Root Cause: When reducing the candidate pool from 128 to 48 experts, the router softmax denominator decreases. To compensate for inflated attention weights, an intermediate implementation applied an attenuation factor of $0.795\times$ to `ffn_down_exps.scale`.

Because this multiplier was applied across 30 consecutive transformer blocks, the attenuation compounded exponentially:

$$(0.795)^{30} \approx 0.00104 \tag{9}$$

The FFN branch's contribution to the residual stream was effectively suppressed by a factor of 1,000. Each block became almost purely an attention passthrough, stripping the network of its factual retrieval and reasoning capacity.

Remedy: Scales must be preserved at exactly $1.0\times$. At runtime, llama.cpp re-evaluates the softmax across only the surviving expert logits, maintaining proper relative dynamic range automatically.

C. Failure Mode 3: DRY Sampling Interference

Symptom: Severe code syntax corruption, broken indentation, and missing identifiers in Python outputs.

Root Cause: The interactive benchmarking harness included a Don't-Repeat-Yourself sampler flag: `--dry-allowed-length 2`, intended to penalize repetitive loops.

However, standard programming syntax relies heavily on 2-character repeating n-grams (`def`, `if`, `in`, `+=`, `==`, `[0`). The DRY sampler severely penalized valid Python keywords, forcing the model to select low-probability corrupted tokens.

Remedy: Removing the DRY sampler and relying on standard nucleus sampling ($\tau = 0.4$, $p = 0.9$, repetition penalty 1.1) restored flawless code syntax.

IV. Experimental Results

A. Experimental Setup

All benchmarks were performed on a consumer laptop workstation:

- **Hardware:** NVIDIA GeForce RTX 4050 Laptop GPU (6141 MiB VRAM), Intel Core i7, 16GB DDR5 system RAM.
- **Inference Engine:** llama.cpp (commit `cbb7d52ec`, CUDA 12).
- **Base Model:** Gemma-4-26B-A4B-it-UD-IQ2_XXS (9.92 GB, 128 experts per layer, 30 layers).
- **Compressed Model:** Gemma-4-26B-A4B-Sub5GB-MAPLE (4.75 GB, 48 experts per layer).

B. Hardware Throughput and Memory Residency

Table I compares execution telemetry between the original 128-expert base model and the 48-expert MAPLE artifact.

TABLE I
System Performance on RTX 4050 6GB Laptop GPU

Model Container	Disk Size	VRAM Alloc	Layers	Speed
Original 128-Exp	9.92 GB	6,141 MiB*	16 / 30	6.9 TPS
MAPLE 48-Exp	**4.75 GB**	**4,180 MiB**	**30 / 30**	**45.4 TPS**
Improvement	**2.09× smaller**	**-32% VRAM**	**100% In-VRAM**	**6.57× faster**

*The original model overflowed VRAM, requiring 14 layers to be paged across PCIe into system RAM.

The $6.57\times$ speedup represents a direct consequence of eliminating PCIe bus transfers. When 100% of model tensors reside in GPU VRAM, token generation runs at full memory bandwidth, delivering 45.4 TPS.

C. Effective Parameter Bitrate

Gemma 4 26B-A4B has 25.6×10^9 total parameters. With the MAPLE container occupying 4,745,987,168 bytes:

$$\text{Effective BPW} = \frac{4{,}745{,}987{,}168 \times 8}{25.6 \times 10^9} \approx \mathbf{1.48} \text{ bits/parameter} \tag{10}$$

This achieves a sub-1.5 BPW effective representation without applying sub-2-bit arithmetic quantization.

D. Task-Level Semantic Fidelity

We evaluated qualitative output fidelity across four benchmark tasks representing the target deployment domain:

- **T1 (Fibonacci Code Synthesis):** Evaluates algorithmic syntax, loop termination, and list accumulation.
- **T2 (Binary Search Tree):** Evaluates class structure, insertion logic, and node traversal.
- **T3 (Rayleigh Scattering):** Evaluates scientific concept explanation and physical equation citation.
- **T4 (Syllogistic Logic):** Evaluates multi-step transitive deduction and formal reasoning.

TABLE II
TASK-LEVEL SEMANTIC KEYWORD FIDELITY

Task ID	Domain	Keywords	Matched	Fidelity
T1: Fibonacci	Code Synthesis	4	4/4	100%
T2: BST Search	Data Structures	4	3/4*	75%
T3: Rayleigh	Physics Deduction	4	4/4	100%
T4: Syllogism	Formal Logic	4	4/4	100%
Overall Mean		**16**	**15/16**	**93.8%**

In Task T2, the partial 75% score occurred solely due to generation truncation at 350 tokens, cutting off the `class Node` definition. Relaxing max tokens to 500 resulted in 100% completion.

E. Out-of-Domain Generalization and Limitations

Evaluating perplexity on a general, out-of-domain validation text (`benchmark_val.txt`) revealed a perplexity increase from 98.9 (original) to 572.4 (MAPLE). This trade-off is fundamental: MAPLE is explicitly a domain-targeted structural distillation. Users requiring broad multilingual or multi-domain support should retain a higher expert budget ($k' \geq 64$). For edge devices where memory is strictly constrained, MAPLE provides an optimal Pareto trade-off.

V. CONCLUSION

MAPLE demonstrates that fine-grained MoE LLMs contain substantial structural redundancy that can be excised post-hoc. By profiling empirical forward-pass activation manifolds, we compress a 26-billion-parameter MoE model to 1.48 effective bits per parameter, unlocking 45.4 TPS inference on commodity 6 GB laptop GPUs with 93.8% domain fidelity.

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